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ORIGINAL RESEARCH

Combined High-Voltage Long-Duration Pulsed Radiofrequency and Ozone Therapy in the Treatment of Pudendal Neuralgia: A Retrospective Case Series

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Purpose: To evaluate the feasibility, safety, and preliminary therapeutic efficacy of combining high-voltage long-duration pulsed radiofrequency with ozone therapy for patients with refractory Pudendal neuralgia (PN).

Patients and Methods: Between January 2019 and December 2024, 18 patients diagnosed with PN based on clinical criteria and diagnostic nerve blocks were included after failing or declining conservative therapies. Following informed consent, all patients underwent a single session of high-voltage, long-duration pulsed radiofrequency treatment combined with ozone therapy, followed by supplementary ozone therapy during hospitalization. Outcomes were assessed using the Visual Analog Scale (VAS), Hamilton Depression Scale (HAMD), Hamilton Anxiety Scale (HAMA), Pittsburgh Sleep Quality Index (PSQI), and SF-36 Health Survey at baseline, 1 week, 4 weeks, 12 weeks, and 24 weeks post-treatment by an independent assessor blinded to clinical details. The primary endpoint was the proportion of patients achieving ≥50% reduction in VAS pain score.

Results: Fifteen patients completed follow-up (three lost to follow-up). At 12 weeks, 93.3% (14/15); achieved the primary endpoint, one patient had persistent pelvic pain but showed improvement in other outcome measures. At 24 weeks, 80.0% (12/15) maintained a favorable response; however, 3 patients experienced symptom recurrence around week 20 and requested retreatment. No serious adverse events were observed during treatment or follow-up periods.

Conclusion: High-voltage, long-duration pulsed radiofrequency combined with ozone therapy showed preliminary efficacy and an acceptable safety profile for patients with refractory pudendal neuralgia. These findings are hypothesis-generating; prospective controlled studies are warranted.

Keywords: pudendal neuralgia, pulsed radiofrequency, ozone therapy, neuropathic pain, case series

Introduction

Pudendal neuralgia (PN) is a debilitating neuropathic pain condition, with an estimated incidence of approximately 1 per 100,000 according to the International Pudendal Neuropathy Foundation, nevertheless, its true prevalence is likely underestimated, with population-based studies suggesting that up to 1% of adults may be affected, and women are more than twice as commonly affected as men, and notoriously difficult to treat. While no overt physical damage is typically observed, the pain is attributed to functional nerve compression or entrapment.[1,2] Diverse triggers such as chronic mechanical stress (eg, cycling), obstetric injury, prior pelvic surgery, or radiation therapy are suspected to initiate a pathological process that culminates in nerve entrapment at the level of the sacrospinous or sacrotuberous

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ligaments.[3] However, PN may also arise secondary to other neuropathic processes, most notably postherpetic neuralgia (PHN) involving the lumbosacral dermatomes, in which clinical features may deviate from the classical Nantes framework while still fulfilling the anatomical and temporal criteria for pudendal-territory neuropathic pain.[3] Herpes zoster in the perineal region is rare, only 5% of cases was reported in the sacral dermatomes.[4] Distinguishing between primary (idiopathic or entrapment) PN and secondary PHN-related pudendal neuropathy is clinically important, as these subtypes differ in pathophysiology, central sensitization mechanisms, and natural history.

Medical ozone formulations generally contain 95–99% O_2 and 1–5% O_3, representing a mixture of these two oxygen allotropes. It has been used as an oxidative medical therapy for over a century and is widely regarded as safe.[5] In 1982, Jacobs et al reported a complication rate of only 0.7 per 100,000 treatments, with most adverse events resulting from improper administration.[6] Ozone therapy exerts moderate oxidative stress and modulates immune and inflammatory responses, contributing to its broad therapeutic potential.[7] It is widely applied via local injections to joints and intervertebral discs for managing arthritis and pain.[7,8] Our previous clinical studies have shown that percutaneous ozone injection is effective and well tolerated in the treatment of trigeminal neuralgia and postherpetic neuralgia.[9]

Pulsed radiofrequency (PRF) is a minimally invasive neuromodulation technique that provides rapid pain relief with few adverse effects. Conventional PRF is typically performed at 45 V for 2 minutes, whereas high-voltage long-duration PRF employs substantially higher voltages (up to 90–99 V) and prolonged stimulation durations (15 minutes).[10] Wang et al modified these parameters to a maximum of 90 V and a total stimulation time of 15 minutes, a protocol termed high-voltage long-duration PRF. Patients were followed for 12 weeks to evaluate its efficacy.[11] In our previous clinical studies, Zhang et al applied an even higher voltage (99 V) combined with ozone therapy for postherpetic neuralgia. After one year of follow-up, this combined high-voltage PRF and ozone treatment achieved superior pain relief compared with ozone therapy alone.[9]

The rationale for combining PRF with ozonated water in PN is grounded in complementary mechanistic pathways. Although ozone therapy has not been previously reported specifically for pudendal neuralgia, its established efficacy in other peripheral neuropathic pain conditions (trigeminal neuralgia and postherpetic neuralgia) supports its mechanistic plausibility in this indication. PRF primarily exerts neuromodulatory effects at the dorsal root ganglion and peripheral nerve level, whereas ozonated water provides localized anti-inflammatory and microcirculatory benefits. In PN—where nerve entrapment, perineural inflammation, and central sensitization often coexist—simultaneous neuromodulation and anti-inflammatory intervention may produce synergistic analgesia.

The optimal management strategy for pudendal neuralgia remains elusive.[10,12] Therefore, more effective and comprehensive integrated treatment strategies, including combination therapies, need to be considered. The primary objective of this study was to determine the proportion of patients achieving at least 50% pain reduction at 12 weeks. Secondary objectives included characterizing changes in mood, sleep quality, and health-related quality of life, and documenting the safety profile of the combined procedure.

Materials and Methods

Setting and Participants

This study was a retrospective analysis of data collected within a standardized prospective observational protocol. Of the 18 patients initially enrolled to receive combined therapy, 3 were lost to follow-up because of failed to contact, yielding a final cohort of 15. Efficacy was evaluated using the VAS, HAMD, HAMA, PSQI and SF-36 at baseline and post-procedure monitoring occurred at 1 week, then again at 4, 12, and 24 weeks. This study was conducted in accordance with the Declaration of Helsinki. Patient identifiers were removed or anonymized prior to analysis to ensure confidentiality. The study received ethical approval from the Institutional Review Boards of Aviation General Hospital of China Medical University (HK-2025-50) and the Institutional Review Boards of Affiliated Hospital of Shandong Second Medical University (SDSMU-2025-KY-241).

The diagnosis of PN was established through systematic application of the Nantes criteria:[13] (1) pain in the anatomical territory of the pudendal nerve; (2) pain exacerbated by sitting; (3) pain not waking the patient at night; (4) no objective sensory loss on clinical examination; and (5) positive diagnostic pudendal nerve block. Patients were required to fulfill criteria 1, 2, and 5 as mandatory items; criteria 3 and 4 were assessed when clinically feasible. Additionally, diagnostic blocks were performed under ultrasound guidance with the patient in lateral decubitus or prone, a low-frequency curved probe traces the hip bone to the Alcock canal entrance. Using an in-plane technique, a 22G needle is inserted from the medial aspect into the canal, with the tip positioned adjacent to the nerve. Location is confirmed by 0.2 mA electrical stimulation without paresthesia. After negative aspiration, 10 mL of 0.5% ropivacaine is slowly injected for diagnostic blockade. A positive response was defined as ≥50% transient pain relief.

Inclusion criteria were: (1) pain in the anatomical territory of the pudendal nerve; (2) age >18 years old; (3) VAS pain scores ≥ 4 points during enrollment; (4) natural course of disease longer than 3 months.

Exclusion criteria were: (1) contraindications to ozone injection, such as fatty liver, acute myocardial infarction, thyroiditis, or other relevant conditions; (2) severe cardiopulmonary disease, hepatic or renal insufficiency, or other life-threatening illnesses; (3) coagulation dysfunction; (4) localized infection at the puncture site; and (5) missed visits during the treatment period. A patient-flow diagram is provided in Figure 1.

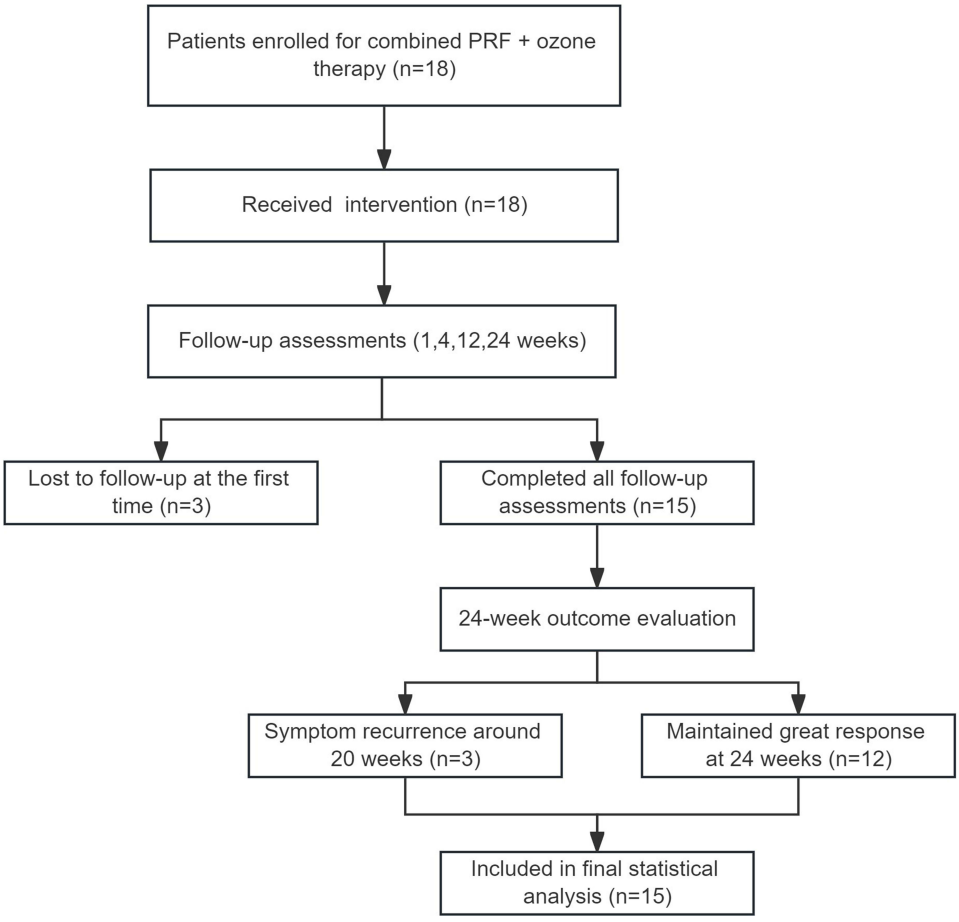


Figure 1 Patient flow diagram.

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PRF

Patients were placed in a prone position with the sacral region exposed. Continuous monitoring of electrocardiogram (ECG), peripheral oxygen saturation (SpO_2), and blood pressure (BP) was maintained throughout the procedure. S2–S4 sacral foramina were identified and marked by CT. After skin preparation with iodine, 2 mL of 1% lidocaine was administered for local anesthesia. The specific sacral nerves to be treated (among S2-S4) were determined for each patient based on their clinical presentation and the distribution of pain and sensory abnormalities. For instance, if a patient's pain was primarily in the perineal region and the pain was relatively localized, then S2 and S3 were targeted. Radiofrequency needles (22G, 5 cm) were inserted into the second through third posterior sacral foramina at depths of 5.0 cm, 5.0 cm respectively, with the needle oriented slightly upward and inward. Intraoperative three-dimensional reconstruction was then used to confirm needle tip placement within the posterior sacral foramen Figure 2A–D. Electrodes were then inserted, and the radiofrequency generator was connected. Sensory testing (50 Hz, 0.4–0.6 V) was performed to elicit abnormal sensations—such as numbness, soreness, or burning—in the pudendal nerve distribution. If no response was obtained, the needle was repositioned until stimulation reproduced the patient's characteristic pain, after which it was fixed in place. Treatment commenced once the patient confirmed a pain sensation similar to their usual discomfort. Pulsed radiofrequency stimulation was then applied at 2 Hz under sedation and analgesia, with a pulse width of 20 ms and a target temperature of 42 °C. The initial stimulation voltage was set at 40 V and gradually increased to the patient's maximum tolerance (up to 99 V in all cases). The total stimulation duration was 15 minutes.

Ozone Therapy

In this case series, ozone was generated by fusing gaseous ozone with sterilized saline using dedicated equipment to produce a defined concentration, which was then injected into the target nerve to allow ozonated water to diffuse around the nerve root. Subsequent to the application of long-duration, high-voltage pulsed radiofrequency therapy, the radiofrequency needle core was gently withdrawn, Ozone therapy was subsequently administered via the same cannula to the exact same nerve(s) that had just undergone PRF treatment. This combined approach was applied to all treated nerves. Subsequently, 5 mL of 0.25% lidocaine and 10 mL of ozonated water through the indwelling cannula, perineural

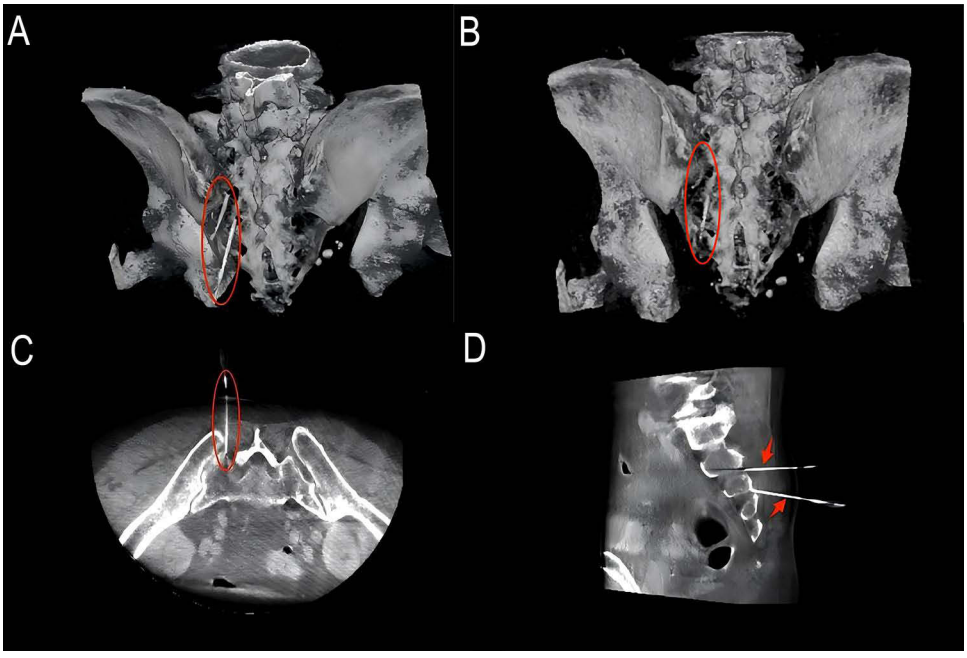


Figure 2 Illustrates a patient who received two-needle therapy. Confirmation that all of the radiofrequency needles are located in the sacral foramen. (**A** and **B**) Intraoperative 3D reconstruction (**C**) Intraoperative transverse section image (**D**) Intraoperative sagittal section image. (Red circles and arrows in (**A**–**D**) indicate the position of the radiofrequency needle.).

infiltration was performed with the solution concentrated at 23 μg/mL. Patients were observed for 30 minutes post-procedure to monitor for adverse reactions before being transferred to their hospital room for further monitoring.

For postoperative ozone therapy alone, treatment was administered once daily on weekdays for 1 to 2 weeks. Using ultrasound guidance, the physician precisely located the pudendal nerve and, under local anesthesia, inserted 22G needle at predetermined site, 2mL 1% Lidocaine and 10 mL of ozonated water (23 μg/mL) was injected. To safeguard against adverse events, every patient remained under monitoring for at least half an hour after each treatment session concluded. All ozonated water used in this study was prepared fresh for immediate use and was not left standing for more than five minutes. Sessions were discontinued when VAS decreased to < 2 or after a maximum of 5 sessions.

Data Collection

After non-invasive treatments—including lifestyle modifications, physiotherapy, and oral medications such as amitriptyline and pregabalin[14]—failed to provide adequate relief, the patient underwent high-voltage pulsed radiofrequency combined with ozone therapy. We collected detailed patient information, including age, gender, disease duration, affected side, pain location, history of comorbidities, VAS scores, mechanical nociceptive sensitization thresholds, number of ozone therapy, and patient self-perception. The primary outcome was the proportion of responders, defined a priori as patients achieving ≥50% reduction in VAS score from baseline. The secondary outcomes measures comprised VAS, HAMD, HAMA, PSQI and SF-36 scores assessed pre-treatment, post-treatment, and at each follow-up interval. Additional outcomes including duration of pain relief and occurrence of complications. All assessments were conducted by a blinded researcher at 1week, 4 weeks, 12weeks, and 24 weeks following treatment. All patients provided written informed consent prior to the interventional procedure. Because the present analysis was conducted retrospectively on existing prospectively collected clinical data, a waiver of additional informed consent for data analysis was granted by the Institutional Review Boards of Aviation General Hospital of China Medical University and the Institutional Review Boards of Affiliated Hospital of Shandong Second Medical University.

Data Analysis

Statistical analyses were performed using Graphpad Prism 10.1.2. Given the small sample size (n = 15) and repeated-measures design, the Friedman test was used to assess overall differences in ordinal or non-normally distributed outcomes across the five assessment timepoints. When the Friedman test yielded a statistically significant result ($p < 0.05$), post-hoc pairwise comparisons between baseline and each follow-up timepoint were conducted using the Wilcoxon signed-rank test, with the significance threshold adjusted via Bonferroni correction ($\alpha = 0.005$ per test, accounting for 10 pairwise comparisons). Data normality was evaluated using the Shapiro–Wilk test. Descriptive statistics are reported as mean ± standard deviation (SD) for normally distributed continuous variables and as median with interquartile range (IQR) for non-normally distributed continuous variables. Categorical variables are summarized as frequencies and corresponding percentages.

Results

This study included 15 patients with PN who underwent combined high-voltage pulsed radiofrequency and ozone therapy. The cohort comprised four males and eleven females (N = 15) with a mean age of 62.7 ± 14.0 years. Disease duration varied widely, ranging from 3 months to 8 years. Five patients had a history of cardiovascular comorbidities, including hypertension and type 2 diabetes mellitus. Two patients reported the presence of anxiety and depressive disorders prior to presentation Additionally, five patients had undergone prior surgical procedures, including repair of anterior vaginal wall prolapse, radical rectal cancer surgery, excision of external hemorrhoids, uterine endometrial cancer surgery, and radical left breast cancer surgery. No patient reported chronic pain outside the pudendal region. A summary of patient demographics and baseline characteristics is presented in Table 1, with special cases described separately.

A pretreatment assessment revealed that all patients experienced PN symptoms characterized by sensations described as burning, numbing, deep, heavy, and etc.[15] Associated symptoms included dyspareunia (n = 1), dyschezia (n = 2), dysuria (n = 9), and pain exacerbated by sitting (n = 6). Most patients reported mood disturbances, including varying degrees of anxiety and depression, on pre-hospitalization questionnaires, likely attributable to chronic pain. The

Table 1 Characteristics of the Patient Population (n = 15 Except Where Indicated)

Variables	Mean (SD) or Frequency (%) N=15
Genders	
Male	4 (26.7)
Female	11 (73.3)
Age, years	62.7 (14)
Weight, kg	65.1 (10.4)
Pain	
Dyschezia	
Yes	2 (13.3)
No	8 (53.3)
Not reported	5 (33.3)
Dysuria	
Yes	9 (60.0)
No	2 (13.3)
Not reported	4 (26.7)
With sitting	
Yes	6 (40.0)
No	2 (13.3)
Not reported	7 (46.7)
Dyspareunia	
Yes	1 (6.7)
No	14 (93.3)
Instigating factor	
PHN	11 (73.3)
Classic PN	4 (27.7)
Physical examination	
Vulvar allodynia	
Yes	9 (60.0)
No	3 (20.0)
Not reported	3 (20.0)
Ischial spine tenderness	
Yes	8 (53.3)
No	4 (26.7)
Not reported	3 (20.0)

presumed precipitating factors for PN in this cohort included prior herpes zoster infection with subsequent postherpetic neuralgia (PHN) involving the pudendal nerve territory in 11 patients, classic PN in 4 patients. In the PHN subgroup, zoster-related pain persisted for at least 3 months after rash resolution and was strictly localized to the pudendal nerve distribution, fulfilling our criteria for secondary PN following PHN. Physical examination findings including sciatic spine tenderness in 8 patients and vulvar allodynia in 9. Prior treatments mainly involved pharmacotherapy, including pregabalin, gabapentin, diclofenac sodium, aminoglutethimide, and ibuprofen. Some patients had also undergone invasive interventions such as nerve blocks, selective posterior rhizotomy, and acupuncture due to inadequate response to medication.

In this study, patients with PN who underwent the combined therapy experienced significant pain relief compared to the preoperative period. The Friedman test revealed a statistically significant overall effect of time on VAS scores across the four assessment timepoints ($p < 0.001$). Post-hoc pairwise comparisons with Bonferroni correction demonstrated that VAS scores were significantly reduced at 1 week, 4 weeks, 12 weeks, and 24 weeks (all $p < 0.001$) compared with baseline. However, one patient complained of minimal relief at the pudendal symphysis and significant relief in the rest

Table 2 Specific Pain Description, Pain Relief After Treatment or Improvement After Treatment, Number of Ozone Therapy per Patient

Patient Number	Age	Primary Diagnosis	Description of Pain	Number of Ozone Injections	VAS				
					Pre-Treatment	1 w	4 w	12 w	24 w
1	78	PHN-related secondary PN	Chronic, stabbing	3	7	1	1	1	2
2	69	Classic Pudendal Neuralgia	Stabbing, deep	3	6	1	0	0	2
3	66	PHN-related secondary PN	Chronic	5	7	2*	2	3	5
4	70	PHN-related secondary PN	Chronic	5	7	1	1	2	3
5	70	PHN-related secondary PN	Chronic, aching	3	8	3	2	2	6
6	70	Classic Pudendal Neuralgia	Burning, numbness	4	7	2	2	2	2
7	30	PHN-related secondary PN	Chronic	4	7	2	1	1	3
8	68	PHN-related secondary PN	Chronic	5	7	2	1	2	3
9	51	PHN-related secondary PN	Deep, sharp	4	6	1	1	2	2
10	79	Classic Pudendal Neuralgia	Stabbing, heavy	5	7	1	1	2	5
11	62	PHN-related secondary PN	Chronic	4	7	1	1	1	2
12	54	Classic Pudendal Neuralgia	Burning, deep	3	6	1	1	1	2
13	75	PHN-related secondary PN	Sharp, radiating	5	5	2	1	1	2
14	61	PHN-related secondary PN	Heavy, sharp	3	8	2	2	1	2
15	38	PHN-related secondary PN	Chronic	5	4	0	0	1	1

Notes: PHN-related secondary PN: Postherpetic Neuralgia (PHN)-related secondary pudendal neuropathy. The table shows the change in VAS scores post-treatment. *After treatment, the patient scored 4–5 at the pubic symphysis and 2 at the rest of the body.

of the area after treatment. This outcome may be due to incorrect targeting of the area or limited treatment effectiveness (Table 2). Additionally, patients demonstrated improved quality of life and enhanced ability to engage in non-sedentary activities such as bicycling. Mood disorders, including anxiety and depression, also showed marked improvement concomitant with pain reduction. There was an upward trend in pain relief and mood disorder improvement over 24 weeks post-treatment (Figure 3). The Friedman test indicated significant overall time effects for HAMD, HAMA, and PSQI scores. Post-hoc analyses confirmed significant improvements at all follow-up timepoints compared with baseline (all $p < 0.001$).

No patients developed infections or neurological complications—such as fecal or urinary incontinence—during treatment or follow-up. Quality of life improvements persisted throughout the follow-up period (Figure 4). The Friedman test demonstrated a significant overall effect on SF-36 scores ($p < 0.001$), with post-hoc comparisons showing significant improvements at 1 week, 4 weeks, 12 weeks, and 24 weeks compared with baseline (all $p < 0.001$). Enabling some patients to resume normal physical activities, including exercise and prolonged sitting.

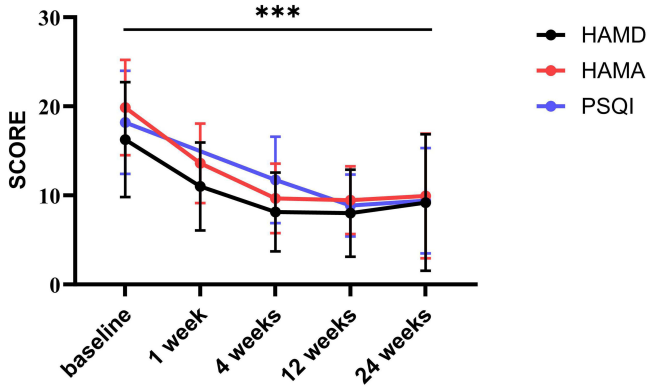


Figure 3 Scores in changes in Hamilton Depression Scale (HAMD), and Hamilton Anxiety Scale (HAMA), Pittsburgh Sleep Quality Index (PSQI) at follow-up points after treatment (Friedman test: all $p < 0.001$*** for overall time effect.).

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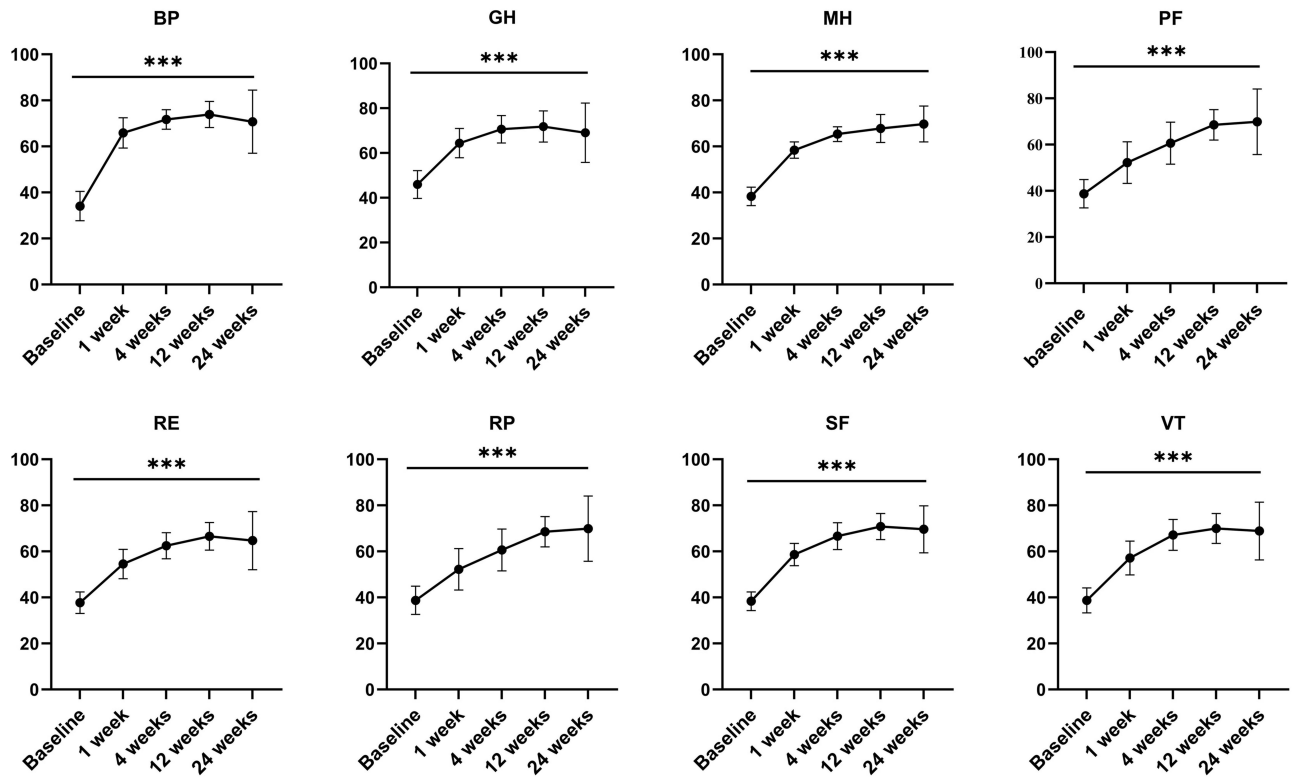


Figure 4 Scores in changes in SF-36 health survey questionnaire (SF-36) at follow-up points after treatment (Friedman test: $p < 0.001$*** for overall time effect.).

Discussion

PN is a complex clinical condition. PN treatment approaches can be broadly categorised as invasive or non-invasive, though each method has limitations and potential side effects. Several patients in this study presented with secondary PN following PHN. In these cases, herpes zoster had previously affected the lumbosacral or perineal dermatomes, and residual neuropathic pain persisted in the anatomical territory of the pudendal nerve for at least 3 months, with other potential causes of pelvic or perineal pain being excluded. Although this secondary subtype occasionally involved more diffuse dysesthesia or sleep disturbance, the core clinical picture remained consistent with pudendal neuralgia and did not contradict the Nantes diagnostic framework. In recent years, PRF and ozone therapy have attracted attention as potential pain management solutions. Multiple studies have confirmed the therapeutic efficacy of these two treatments. Our team has previously demonstrated that this combination therapy produces favourable outcomes for postherpetic and trigeminal neuralgia.[9] This study sought to determine how effective this combined therapeutic approach proves in treating PN.

A critical limitation of this study is the inclusion of patients with both classic entrapment-type PN and secondary PHN-related pudendal neuropathy. The Nantes criteria were developed for primary pudendal nerve entrapment and emphasize pain exacerbated by sitting, absence of night-time pain waking the patient, and no objective sensory loss.[12] While our PHN subgroup fulfilled the anatomical criterion (pain in the pudendal territory) and responded positively to diagnostic nerve block, these patients differed from the classic entrapment cohort in pathophysiology: PHN involves persistent viral-induced inflammation, dorsal root ganglion damage, and pronounced central sensitization, whereas classic PN is primarily a peripheral entrapment syndrome.[2] This heterogeneity may have influenced treatment response patterns. For instance, PHN-related PN may be more responsive to anti-inflammatory interventions such as ozonated water, whereas entrapment-type PN may benefit more from the neuromodulatory effects of PRF. Future studies should stratify patients by etiology and report outcomes separately.

With a follow-up window of similar duration, Fang et al's randomized trial achieved 92.1% success—almost identical to the 93.3% effectiveness rate seen here at 12 weeks. Their study population comprised 77 individuals with PN,

randomly assigned such that 38 received the PRF–PNB combination and 39 received PNB as standalone therapy. At 12 weeks, the clinical effectiveness rate was 92.1% in the PRF group compared to 35.9% in the PNB group. Fang et al also reported a lower complication rate associated with PRF in PN treatment.[16] Consistent with these findings, no serious adverse events were observed in either the PRF or nerve block groups, paralleling the safety profile demonstrated in our case series but the result should be interpreted with caution. Fang et al also reported a lower complication rate associated with PRF in PN treatment. Consistent with these findings, the safety profile also demonstrated in our case series.[16]

The precise mechanism of action of PRF remains unclear. PRF operates differently from continuous radiofrequency (CRF): rather than a steady current, it administers high-intensity bursts with built-in pauses. These latency intervals permit heat dissipation, keeping tissue temperatures under 42°C and sparing the nerve from structural injury.[17] Hamann et al showed that PRF preferentially acts on neurons bearing small-diameter axons, particularly Aδ and C nociceptive fibers.[18] Consequently, patients exhibited no complications or motor deficits during long-term follow-up, and the treatment could be safely repeated in cases of pain recurrence.[14] In contrast to the protocol described by Wang et al, which employed a maximum voltage of 90 V over 15 minutes, our study utilized a slightly higher maximum voltage of 99 V for the same duration of continuous stimulation.[11] Importantly, no nerve damage was observed, supporting the safety and repeatability of our treatment regimen. Moreover, compared to other invasive approaches such as PNB, our protocol incorporates intraoperative three-dimensional reconstruction to achieve more precise needle localization and substitutes local anesthesia with ozonated water, which is associated with fewer side effects, thereby further minimizing the risk of nerve injury.

The therapeutic use of ozone stretches back over a century, spanning numerous disease categories.[5] It has demonstrated anti-inflammatory and anti-infective properties, as well as accelerating blood metabolism and regulating immune function.[7] While there are currently no studies on the synergistic mechanisms of this combined therapy, it is possible to hypothesise that ozone therapy's analgesic effects may be achieved by activating the descending pain-blocking system and promoting endorphin release.[7] Combining this with neuromodulation could enhance therapeutic efficacy. To verify the safety and efficacy of higher concentrations, this study used ozonated water at twice the concentration (23 μg/mL).[19–21] Numerous studies have demonstrated at concentrations of ≤30 μg/mL, ozone effectively removes pain-inducing factors and perineural adhesions surrounding the dorsal root ganglion (DRG), spinal nerve roots, and sympathetic ganglia, ameliorates local hypoxia, and facilitates the repair of damaged nerves. Concurrently, it stimulates vascular endothelial cells to release nitric oxide (NO) and platelet-derived growth factor (PDGF), thereby inducing vasodilation, enhancing local microcirculation, and accelerating the resolution of inflammation.[22] Compared to previous studies, this concentration also falls within the safe range.

This case series represents the first documentation of the combined therapy for patients with PN. The effectiveness of the treatment was evaluated using VAS, HAMD, HAMA, PSQI and SF-36. However, the absence of a placebo control group limits the conclusiveness of these findings. While the inclusion of a control group was considered, ethical concerns regarding the conduct of randomized controlled trials (RCTs) involving sham procedures in this patient population were deemed significant. Van Zundert et al have highlighted the ethical and methodological challenges associated with sham-controlled RCTs in interventional pain management.[23]

This combined therapeutic approach was applied specifically to patients unresponsive to non-invasive treatments or PNBs, with the aim of providing prompt pain relief and reducing psychological and physical distress. Future studies with larger sample sizes are warranted to further validate the efficacy of this method. Additionally, exploring varying concentrations of ozonated water within safe limits may help optimize and prolong treatment effects. Currently, this protocol demonstrates a success rate, sustained analgesic effect, and an excellent safety profile, with no adverse events reported. Therefore, it may be considered a promising treatment option for patients with PN.

This study has several limitations that must be acknowledged. The retrospective uncontrolled design precludes any inference of causality or attribution of efficacy to either PRF or ozone alone. Placebo effects, natural disease fluctuation, and regression to the mean cannot be excluded. The sample size is small, and the 24-week follow-up duration is insufficient to assess long-term durability in a chronic condition characterized by recurrence. Cohort is etiologically heterogeneous, with 73.3% of patients having PHN-related PN, which may differ in treatment responsiveness from classic PN. Imaging documentation lacks representative ultrasound images, and the dual-modality guidance protocol

requires further standardization. Number of supplementary ozone sessions varied between patients (3–5), introducing a potential source of treatment heterogeneity. Statistical analysis was limited by the small sample size; although we applied non-parametric repeated-measures methods and Bonferroni correction, the power to detect subtle differences or to perform meaningful subgroup analyses was low. The ethics approval was granted retrospectively after the initiation of data collection, reflecting the transition from a clinical quality-improvement protocol to a formal research analysis; this is a common scenario in retrospective clinical studies but requires transparent disclosure. Finally, the absence of a patient-blinded control group and the inability to blind the interventionalist are inherent limitations of the design.

Conclusion

These findings suggest that high-voltage long-duration pulsed radiofrequency combined with ozone therapy is an effective, minimally invasive treatment modality with minimal side effects and a favorable safety and feasibility profile. It should be considered preliminary and hypothesis-generating.

Data Sharing Statement

The data that support the findings of this study are available on request from the corresponding author. The data are not publicly available due to privacy or ethical restrictions.

Ethics Approval Statement

This study received ethical approval from the Institutional Review Boards of Aviation General Hospital of China Medical University (HK-2025-50) and the Institutional Review Boards of Affiliated Hospital of Shandong Second Medical University (SDSMU-2025-KY-241).

Patient Consent Statement

As this was a retrospective study, an exemption was received, and there were no patient identifiers, consent was not required for this study.

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Disclosure

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